CocoConnect

A Unified Digital Ecosystem for the Coconut Industry

Requirement Analysis Document

Client: Coconut Growers Association of Sri Lanka (CGASL)

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1. Introduction

1.1 Purpose

This document analyses the requirements gathered for CocoConnect from the Coconut Growers Association of Sri Lanka and estate owners. It identifies the stakeholders and their needs, examines how the business currently works and where it fails, and turns those findings into prioritised, traceable requirements for the system. It also records conflicts and open questions between the sources, so they can be resolved with the client before design starts.

1.2 Scope of the system

CocoConnect is a mobile-first platform that brings the coconut industry's buying, selling, hiring and estate activities into one system. It covers user management, a product marketplace, a workforce and service marketplace, estate and harvest management, ratings and reviews, notifications and administration. It starts with coconut products and is designed to extend to other agricultural products later.

1.3 Sources analysed

Key	Source	Used for
RG	Requirements gathered – notes from the sessions with Mr. Shakila Wijewardena (CGASL)	Business workflows, domain knowledge, stakeholder concerns
RSR	Refined Stakeholder Requirements Document	Stakeholder list, process issues, FR-01 to FR-44, NFR-01 to NFR-08
SRS	Software Requirements Specification, Version 0.1 (Initial Draft)	Scope, user roles, FR01 to FR35, business rules, assumptions, constraints
PROP	Industry Project Proposal	Problem statement, objectives, modules, technology stack, risks

1.4 Method

- Requirement-gathering sessions with the client and estate representatives, recorded as session notes.
- Analysis of the current (as-is) business process to find the problems the system must solve.
- Stakeholder analysis: who uses or is affected by the system, and what each group needs.
- Consolidation of requirements from all sources, removing duplicates and tracing each one back to its source.
- Prioritisation using MoSCoW (Must, Should, Could, and Future for out-of-scope items).
- Gap and conflict analysis between the sources.

2. Problem Analysis

The coconut industry in Sri Lanka depends on phone calls, manual negotiation and separate tools. The sessions with the client showed that the problems fall into four areas.

Area	Current situation	Effect
Product selling	No central marketplace. Prices are negotiated manually, buyer by buyer. Buyers cannot see what is available or in what quantity.	Lack of price transparency; sellers depend on the buyers they already know.
Workforce	Pluckers, huskers, fertilizing and loading workers and tractor suppliers are found through personal contacts. There is no way to check a worker's experience or reliability.	Difficulty finding skilled workers when needed; the client asked specifically for a rating system for workers.
Harvest records	An estate manager enters coconut counts into a Google Sheet; one person counts the nuts for each tree.	Time-consuming data entry, no history or analytics, risk of errors.
Trust and payment	Payment is by cheque or bank transfer. Trust issues exist between buyers and sellers and when hiring workers.	Deals are slow; huskers only start husking after a price has been agreed with the buyer.

3. Stakeholder Analysis

Stakeholder	Role in the system	Main needs and concerns	Influence
Estate Owner	Lists products, hires workers, manages estates	Reach more buyers, get fair prices, find reliable workers, see harvest performance	High
Estate Manager	Runs daily estate work, records harvests	Fast harvest entry (often offline), manage only assigned estates	Medium
Buyer	Finds and buys coconut products	See available stock and quantity, compare prices, trust the seller, contact sellers easily	High
Seller	Lists and manages products	Simple listing with price, quantity, location and selling method	High
Workers and service providers (pluckers, huskers, fertilizing, gathering, loading workers, tractor and equipment suppliers)	Offer services and accept jobs	Receive job requests, set charging method (per tree or per day), build a reputation	Medium
Administrator	Verifies and manages users and listings	Verify accounts, remove fraudulent listings, act on reports	High
CGASL (client)	Sponsors and validates requirements	A platform that serves the wider industry and can grow beyond coconuts	High
Secondary: future product sellers, tourism operators, logistics providers	Possible future users	Selling pepper, fruit and poultry; estate cabana bookings; transport	Low

4. Domain Knowledge

The sessions captured facts about the coconut business that the system must model correctly:

Topic	Finding	Impact on requirements
Coconut types	DC coconut (usually kept for about 21 days before selling) and curry coconut (fresh nut)	Product categories must include both types and coconut husk
Selling method	By kilogram or per nut; with the husk or without the husk; husks are also sold separately	Listings need a unit (kg or nut) and a husk option
Plucking	By pole (kekka) or by climbing; charged per tree or per day	Worker profiles need a service type and a charging method
Quality	Judged by average nut weight (about 615 g), kernel thickness and estate location	Listings should show location and allow quality details
Yield	As the nut count of a tree rises, nut weight falls. Ideal density is 64 trees per acre with a 26-foot gap	Estate records need acreage and tree count
Payment	Cheques and bank transfers today	Online payment is a future enhancement; the first version records deals only
Language	The platform should support the three main languages	English, Sinhala and Tamil interface

5. Functional Requirements

Requirements from all sources were merged and given one ID each. The Source column shows where each requirement comes from, so it can be traced back.

5.1 User management

ID	Requirement	Source	Priority
UM-01	Users shall register an account and select their role.	RSR FR-01, FR-03; SRS FR01	**Must**
UM-02	Users shall log in securely and log out.	RSR FR-02; SRS FR02, FR03	**Must**
UM-03	The system shall give each role access only to its own functions.	RSR FR-03; SRS FR06	**Must**
UM-04	Users shall update their profile information.	RSR FR-04; SRS FR04	**Must**

ID	Requirement	Source	Priority
UM-05	Users shall reset a forgotten password.	SRS FR05	**Should**
UM-06	Administrators shall verify user accounts.	RSR FR-05, FR-40	**Should**

5.2 Product marketplace

ID	Requirement	Source	Priority
PM-01	Sellers shall create product listings with product type, price, quantity, location and selling method.	RSR FR-06, FR-13; SRS FR07	**Must**
PM-02	Sellers shall edit and remove their listings and manage inventory quantities.	RSR FR-09; SRS FR08, FR09	**Must**
PM-03	Buyers shall browse available products and view product details.	RSR FR-07; SRS FR10, FR12	**Must**
PM-04	Buyers shall search and filter products.	RSR FR-08; SRS FR11	**Must**
PM-05	The system shall support selling per kilogram and per nut.	RSR FR-11; RG	**Must**
PM-06	The system shall support the categories DC coconut, curry coconut and coconut husk.	RSR FR-12; SRS 3.2; RG	**Must**
PM-07	Buyers shall contact sellers and agree a price inside the system.	RSR FR-10; SRS FR13; RG (price agreed before husking)	**Must**
PM-08	Buyers shall see seller ratings.	RSR FR-14	**Should**

5.3 Workforce and service marketplace

ID	Requirement	Source	Priority
WM-01	Workers and service providers shall create service profiles with a service category.	RSR FR-15, FR-17; SRS FR14	**Must**
WM-02	Service categories shall include plucking, husking, fertilizing, gathering, loading, tractor services and equipment supply.	RSR FR-17; SRS 3.3; RG	**Must**
WM-03	Workers shall state their charging method (per tree or per day).	RSR FR-18; RG	**Must**
WM-04	Estate owners shall search for workers and service providers.	RSR FR-16; SRS FR15	**Must**
WM-05	Estate owners shall send job requests; workers shall accept or reject them.	RSR FR-19, FR-20; SRS FR16, FR17	**Must**
WM-06	The system shall track the status of each job request.	SRS FR18	**Must**
WM-07	Workers shall be found by distance from the estate.	PROP C3 module 5 (location-based service marketplace)	**Should**

5.4 Estate and harvest management

ID	Requirement	Source	Priority
EH-01	Estate owners shall register estates and update estate information (location, acreage, number of trees).	RSR FR-35 to FR-37; SRS FR19, FR20, FR22	**Must**
EH-02	Authorised users shall view estate information.	SRS FR21	**Must**
EH-03	Estate managers shall record harvest date, tree count, coconut count and total harvest.	RSR FR-23, FR-24; SRS FR23	**Must**
EH-04	The system shall keep harvest history and show harvest reports to estate owners.	RSR FR-25, FR-26, FR-38; SRS FR24, FR25	**Must**
EH-05	Harvest recording shall work offline and synchronise when connected.	RSR FR-27, FR-28; PROP objective 5	**Should**
EH-06	Estate owners shall monitor workforce activity on their estates.	RSR FR-39	**Could**

5.5 Trust, ratings and reviews

ID	Requirement	Source	Priority
TR-01	Buyers and sellers shall rate each other after a completed transaction.	RSR FR-29, FR-30; SRS FR26	**Should**
TR-02	Estate owners and workers shall rate each other after a completed job.	RSR FR-21, FR-22, FR-31, FR-32; RG	**Should**
TR-03	Users shall write reviews, and ratings shall form a public reputation score.	RSR FR-33, FR-34; SRS FR27, FR28	**Should**

5.6 Notifications

ID	Requirement	Source	Priority
NT-01	The system shall notify users about important activity, including job requests and accepted or rejected requests.	SRS FR29 to FR31	**Should**

5.7 Administration

ID	Requirement	Source	Priority
AD-01	Administrators shall manage user accounts and permissions, and suspend users when necessary.	RSR FR-41, FR-44; SRS FR32, FR35	**Must**
AD-02	Administrators shall monitor listings and remove fraudulent ones.	RSR FR-43; SRS FR33	**Must**
AD-03	Administrators shall review reported content and monitor platform activity.	RSR FR-42; SRS FR34	**Should**

6. Non-Functional Requirements

ID	Category	Requirement	Source
NF-01	Platform	The system shall run on Android and iOS mobile devices, with web access for administration.	RSR NFR-01; SRS 2.4
NF-02	Performance	Responses shall take less than 3 seconds under normal conditions.	RSR NFR-03; SRS 4
NF-03	Security	Passwords shall be stored encrypted (hashed), and personal information protected.	RSR NFR-04; SRS 2.6, 4
NF-04	Security	Access shall be authorised by role.	RSR NFR-05; SRS 2.6
NF-05	Availability	The service shall be available 99% of the time.	RSR NFR-06
NF-06	Offline use	Harvest recording shall work without a connection.	RSR NFR-02
NF-07	Usability	The interface shall be simple for users with limited digital experience.	SRS 4; PROP risk 2
NF-08	Language	The interface shall support English, Sinhala and Tamil.	RSR NFR-08; RG
NF-09	Extensibility	New agricultural products and features shall be addable without redesign.	RSR NFR-07; SRS 2.6, 4
NF-10	Maintainability	Source code shall be organised and maintainable.	SRS 4

7. Business Rules

- Only registered users can use protected features, and only the functions allowed for their role.
- A product listing must include the required details: type, price, quantity, location and selling method.
- A worker must choose a service category when creating a service profile.
- A job is only assigned once the worker has accepted the request.
- Ratings can only be given after a completed transaction or service.
- Estate managers can only manage the estates assigned to them.
- Administrators can act against accounts that break platform rules.
- A price is agreed between buyer and seller before work such as husking begins.

8. Gaps and Conflicts Between Sources

Comparing the sources showed points that need a decision with the client:

#	Issue	Sources	Recommendation
1	Multilingual support is required by the stakeholders but listed only as a future consideration in the SRS draft.	RG, RSR NFR-08 vs SRS 6	Treat English, Sinhala and Tamil as a core requirement (NF-08).
2	Location-based services are a core module in the proposal but listed as a future consideration ('GPS based services') in the SRS draft.	PROP C3, D1 vs SRS 6	Keep location-based search in scope (WM-07); defer advanced GPS tracking.
3	The SRS draft assumes most functions need an internet connection, while the stakeholder requirements ask for offline harvest recording.	SRS 2.5 vs RSR FR-27, NFR-02	Make only harvest recording offline-capable (EH-05).
4	The refined requirements list ten stakeholder roles; the SRS draft groups them into six user roles.	RSR 2 vs SRS 2.3	Use the six SRS roles in the system; worker types become service categories.
5	Equipment rental appears in the stakeholder workforce list (tractor and equipment suppliers) and also as a future feature in the SRS draft.	RG, RSR FR-17 vs SRS 6	Include tractor and equipment suppliers as service categories now; full rental booking later.
6	Password reset and account verification each appear in only one source.	SRS FR05; RSR FR-05	Keep both, with Should priority.
7	Payment terms and buyer–seller trust are raised by the client, but no source defines how a deal is confirmed or what happens if it fails.	RG	Clarify with the client: record agreed price and quantity, and handle disputes through administrators.

9. Scope

9.1 In scope

- User management with role-based access
- Product marketplace for coconut products, with in-system contact and price agreement
- Workforce and service marketplace with job requests
- Estate and harvest management, with offline harvest recording
- Ratings, reviews and notifications
- Administration

9.2 Out of scope (future enhancements)

Enhancement	Source
Other products: pepper, mango, lemon, orange, pineapple, poultry	RG; RSR 5
Estate tourism: cabana booking and estate experiences	RG; RSR 5; SRS 6
Online and escrow payments	RSR 5; SRS 6
Yield prediction, productivity and price trend analytics	RSR 5; SRS 6
Weather information and AI-based recommendations	SRS 6

10. Assumptions, Constraints and Risks

10.1 Assumptions

- Users have basic smartphone skills and provide correct information when registering and listing.
- Most functions need an internet connection, except harvest recording.

10.2 Constraints

- Technology stack: Flutter (mobile), Node.js with Express.js (backend), PostgreSQL with PostGIS (database), JWT authentication.
- User data must be protected, and each role may only access its own functions.

- The design must allow new features and products to be added later.

10.3 Risks

Risk	Likelihood	Mitigation
Weak GPS signal in rural areas	High	Offline caching; synchronise when connected
Low digital literacy among growers	High	Local language support and user training
Payment gateway integration challenges	Medium	Trusted local payment APIs, phased integration
Incorrect harvest data entries	Medium	Approval workflows and validation checks
Scope growing beyond project goals	Medium	Limit Phase 1 features; defer advanced functions
Third-party API downtime	Medium	Fallback services and local caching

Source: Industry Project Proposal, section G.

11. Traceability: Problems to Requirements

Problem found in analysis	Requirements that address it
No central marketplace; limited visibility of stock	PM-01 to PM-06
Prices negotiated manually; price agreed before husking	PM-07, business rule 8
Hard to find skilled workers	WM-01 to WM-07
No way to judge worker or trading partner reliability	TR-01 to TR-03, PM-08, UM-06
Harvest counts kept manually in Google Sheets	EH-03 to EH-05
Fraud and untrusted accounts	UM-06, AD-01 to AD-03
Users with limited digital experience, three languages	NF-07, NF-08